

Part 2a: no UV — but the plus prescription needs two distributions per end

Every step, and the one place where a single $[1/\eta]_+$ would be illegitimate

$$\eta = \frac{p^+}{k^+}, \bar{\eta} = 1 - \eta, L = \log \frac{k^+}{\Lambda}, \mathcal{W} = \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2}, \text{phase} = \exp[-ik \cdot (w' - \bar{\eta}z - \eta x)]$$

No, 2a has no ultraviolet divergence. There is nothing to regularise: keep $d_\perp = 2$ everywhere, no $\overline{\text{MS}}$, no counterterm. **But the p^+ integration is not the simple two-sided plus prescription I described before.** 2a has a $\frac{1}{z^2}$ tail *and* a mirror $\frac{1}{x^2}$ tail, each cut off only by its own oscillation — $e^{i\bar{\eta}k \cdot z}$ and $e^{i\eta k \cdot x}$. As $\eta \rightarrow 1$ the first switches off, as $\eta \rightarrow 0$ the second does, so the p^+ integrand behaves as $\frac{A \log(1/\text{pole}) + B}{\text{pole}}$ at *both* ends. A single $[\frac{1}{\eta}]_+$ is then illegitimate, because its $g(0)$ is itself logarithmically divergent. You need **four** distributions, two per endpoint, and the result carries $\log^2 \frac{k^+}{\Lambda}$.

1. There is no UV divergence at $z \rightarrow x$

The reason is in the colour factor, before any numerics. Using the fusion identity at coincident points,

$$\mathcal{C}_{2a}^{af} = -f^{adc} f^{fbe} U^{db}(x) U^{ce}(z) + N_c U^{af}(x) = -f^{adc} f^{fbe} U^{db}(x) [U^{ce}(z) - U^{ce}(x)] = \frac{N_c}{2} [U^{af}(x) - U^{af}(z)] + O((z-x)^2),$$

so $\mathcal{C}_{2a} = O(|z-x|)$. Two powers are gained, not one:

1. the linear vanishing beats the kernel's $\frac{1}{(z-x)^2}$ once $d^2 z = |u| d|u| d\phi$ is counted;
2. the linear piece is **odd** in $\widehat{z-x}$, while the UV residue $C_{UV}(\eta) \mathcal{W} \frac{1}{(z-x)^2}$ is **even**, so the angular average removes it too.

Measured: $u^2 \times \langle \text{integrand} \rangle_{\hat{u}}$ as $u = z - x \rightarrow 0$ (a constant means log divergent):

$ z-x $	2a, with colour	colour stripped
10^{-2}	2.107×10^{-5}	0.01348138
10^{-3}	2.108×10^{-7}	0.01348135
10^{-4}	2.108×10^{-9}	0.01348135
10^{-5}	2.108×10^{-11}	0.01348135

The colour-stripped column is the constant that 2b has to regularise; the 2a column falls as $|z-x|^2$. **Nothing to handle.**

2. What 2a does have: two long-distance tails

Both z and x are integration variables over the whole plane (each carries a Wilson line), and the colour-stripped integrand is invariant under $(x \leftrightarrow z, \eta \leftrightarrow \bar{\eta})$ — verified earlier to 1.6×10^{-13} . So a tail in one implies a tail in the other.

A diagnostic trap I fell into. With the phase kept, the angular average of a $\frac{1}{v^2}$ tail is $\sim \frac{J_0(q|v|)}{v^2}$, which oscillates and decays — so “ $|v|^2 \langle f \rangle \rightarrow \text{constant}$ ” is the *wrong* test and wrongly says there is no tail. Measure the tail coefficient with the oscillation stripped, then restore it analytically.

Tail coefficients, phase stripped ($|v|^2 \langle f \rangle \rightarrow \text{constant}$):

$ v $	$z \rightarrow \infty, \eta = 0.4$	$x \rightarrow \infty, \eta = 0.6$
10^2	0.00676818	-0.11051726
10^3	0.00676868	-0.11052668
10^5	0.00676868	-0.11052678

The z coefficient matches the closed form $\mathcal{V}_z(\eta) = \frac{\mathcal{W}}{2k^+} \left[\frac{1}{d_\perp} \frac{\bar{\eta}}{\eta} + \frac{1}{\bar{\eta}} \right]$ at every η :

η	0.20	0.40	0.60	0.80	0.95
measured	0.00910271	0.00676868	0.00793570	0.01435428	0.05609041
closed form	0.00910271	0.00676868	0.00793570	0.01435428	0.05609041

and $\mathcal{V}_z \rightarrow \frac{\mathcal{W}}{2k^+\bar{\eta}}$ as $\eta \rightarrow 1$; by the mirror symmetry $\mathcal{V}_x \rightarrow \frac{\mathcal{W}'}{2k^+\eta}$ as $\eta \rightarrow 0$. Neither tail is switched off by $2a$'s colour factor: $\|\mathcal{C}_{2a}\|$ is 1.30 at $U(z) \rightarrow 1$ and 2.53 at $U(x) \rightarrow 1$.

The oscillation is the only cut-off

$$\int_{|z|>R_0} d^2z \frac{e^{i\bar{\eta}k \cdot z}}{z^2} = 2\pi \int_{R_0}^{\infty} \frac{dr}{r} J_0(\bar{\eta}k_{\perp}r) = -2\pi \log \left[\frac{\bar{\eta}k_{\perp}R_0 e^{\gamma_E}}{2} \right] = 2\pi \log \frac{1}{\bar{\eta}} + \text{const}$$

$\bar{\eta}k_{\perp}$	tail integral	$-2\pi \log \frac{q e^{\gamma_E}}{2}$
10^{-1}	15.2038395	15.195988
10^{-2}	29.6636354	29.6635568
10^{-3}	44.1311265	44.1311257

Only the isotropic part of the tail produces this: the anisotropic $\frac{\hat{z}^m \hat{z}^n}{z^2}$ piece transforms with J_2 , and $\int \frac{dr}{r} J_2(qr)$ has no small- q logarithm.

3. Therefore the p^+ integrand is $\frac{A \log + B}{\text{pole}}$

Combining the tail coefficient's $\frac{1}{\bar{\eta}}$ with the tail integral's $\log \frac{1}{\bar{\eta}}$, and the same on the x side, the z - and x -integrated function $G(\eta) = \int_{x,y,z,x'} (\dots)$ behaves as

$$G(\eta) \xrightarrow{\eta \rightarrow 0} \frac{A_0 \log \frac{1}{\eta} + B_0}{\eta}, \quad G(\eta) \xrightarrow{\eta \rightarrow 1} \frac{A_1 \log \frac{1}{\bar{\eta}} + B_1}{\bar{\eta}},$$

$$A_1 = \frac{\pi}{k^+} \mathcal{W} \left[-f^{adc} f^{fbc} U^{db}(x) + N_c U^{af}(x) \right], \quad A_0 = \text{the same with } x \leftrightarrow z, \quad U(x) \rightarrow 1.$$

This is why a single plus prescription will not do. The prescription $\int \frac{d\eta}{\eta} [g(\eta) - g(0)]$ requires $g(0)$ **finite**, and here $g(\eta) = \eta G(\eta) = A_0 \log \frac{1}{\eta} + B_0$ diverges at $\eta = 0$. Writing $\left[\frac{1}{\eta}\right]_+$ alone and putting $\log \frac{k^+}{\Lambda}$ on the δ would be subtracting infinity from infinity, and the finite part would be meaningless. *This corrects the two-sided-but-single-distribution form I gave you earlier*, which assumed $h(\eta)$ stood in for the whole η -dependence — it does for the integrand at fixed z , but not after the z and x integrals are done, which is where the extra log is generated.

4. The correct treatment: two distributions at each end

$$\frac{1}{\eta} = \left[\frac{1}{\eta}\right]_+ + \delta(\eta) \log \frac{k^+}{\Lambda}, \quad \frac{\log \frac{1}{\eta}}{\eta} = \left[\frac{\log \frac{1}{\eta}}{\eta}\right]_+ + \frac{1}{2} \delta(\eta) \log^2 \frac{k^+}{\Lambda},$$

and the same two in $\bar{\eta}$ — four in all. Both identities checked against the direct cut-off integral with a smooth test function:

Λ/k^+	$\frac{1}{\eta}$ direct	$\frac{1}{\eta}$ plus form	$\frac{\log(1/\eta)}{\eta}$ direct	plus form
10^{-3}	5.8808211927	5.8795214927	22.7101373625	22.6998595028
10^{-6}	12.7872780717	12.7872767717	94.2755032544	94.2754839943
10^{-9}	19.6950320520	19.6950320507	213.5681915083	213.5681914800

5. The step-by-step recipe for $2a$

- Set $d_{\perp} = 2$** and drop all thought of dimensional regularisation. $2a$ is UV finite.
- Do the transverse integrals first**, at fixed η : $\int_{x,y,z,x'}$ with the phase $\exp[-ik \cdot (w' - \bar{\eta}z - \eta x)]$ and the colour factor $-f^{adc} f^{fbc} U^{db}(x) [U^{ce}(z) - U^{ce}(x)]$. Call the result $G(\eta)$. Both integrals converge — $z \rightarrow x$ by §1, large $|z|$ and $|x|$ by the oscillations.
- Extract the four endpoint constants A_0, B_0, A_1, B_1** . A_0 and A_1 you have in closed form above; B_0 and B_1 come out of the same transverse integrals with the tail removed.
- Subtract and integrate:**

$$\int_{\Lambda}^{k^+-\Lambda} \frac{dp^+}{2\pi} G = \frac{k^+}{2\pi} \left\{ \frac{A_0 + A_1}{2} L^2 + (B_0 + B_1)L + \int_0^1 d\eta \left[G - \frac{A_0 \log \frac{1}{\eta} + B_0}{\eta} - \frac{A_1 \log \frac{1}{\bar{\eta}} + B_1}{\bar{\eta}} \right] \right\} + O(\Lambda)$$

The bracket is integrable on $[0, 1]$ by construction and is the numerical piece.

Step 4 is the plus prescription, written so that you can see what it does: the two δ coefficients are L and $\frac{L^2}{2}$, and the remainder is the $[\]_+$ part.

6. Where this leaves the row

	UV	p^+ treatment	rapidity logs
2a	none	four distributions, two per endpoint	$\frac{A_0+A_1}{2} \log^2 \frac{k^+}{\Lambda} + (B_0 + B_1) \log \frac{k^+}{\Lambda}$
2b	$\frac{1}{\epsilon}$, coefficient $2L - \frac{11}{6}$	$T_3+T_5+T_6$ plain; T_1, T_2, T_4 subtract then one-sided plus	$-\log^2 \frac{k^+}{\Lambda}$ from I_1 , plus $2L$ on the pole

Both halves carry a $\log^2 \frac{k^+}{\Lambda}$, which is what you should expect of a self-energy loop on the measured gluon: it is the Sudakov double logarithm. Their sum, together with the companion rows, is what has to reduce to the JIMWLK structure. Worth checking once you have the companions: the $\log^2$ coefficients should combine into something proportional to the LO cross section, since a double rapidity log cannot survive in a single-log evolution equation.

Checks: `src/rowB2_2a_full.py` — §1 the UV scan with a smooth $U(z)$ field; §2 the two tail coefficients with the phase stripped, the closed form for $\mathcal{V}_z$ at five η , the Bessel tail integral, and the two colour limits; §4 the two distribution identities against direct quadrature. Output in `RESULTS_rowB2_2a_full.txt`.